The Chinese University of Hong Kong, Shenzhen
SDS · School of Data Science

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MAT 3007 — Optimization

Exercise Sheet 1

Problem 1 (Point Clouds and Rectangles): (approx. *20* pts)
Let $\boldsymbol{y}^1, \boldsymbol{y}^2, \ldots, \boldsymbol{y}^k$ be k given points in $\mathbb{R}^2$, i.e., it holds that $\boldsymbol{y}^i \in \mathbb{R}^2$ for all $i = 1, \ldots, k$.

a) We want to find a rectangle in $\mathbb{R}^2$ with *minimum area* that contains all of the different points $\boldsymbol{y}^1, \ldots, \boldsymbol{y}^k$. Formulate this task as an optimization problem and classify the type of your modeled minimization problem.

b) Suppose now that we want to find a rectangle with *minimum perimeter* enclosing the points $\boldsymbol{y}^1, \ldots, \boldsymbol{y}^k$. Explain how this changes the formulation of the optimization problem.

Can you give a solution to this problem? Will such a solution be different from a solution to the minimum area problem discussed in part a)? Explain your answer!

Hint: You can assume that the rectangle is aligned with (i.e., parallel to) the x- and y-axis. Notice that such rectangles can be determined by one of their corner points and by their length and width.

Problem 2 (Production Planning): (approx. *20* pts)
A company produces three kinds of products. The product of type I requires $\frac{1}{3}$ hours of assembly labor, $\frac{1}{5}$ hours of testing, and 13 RMB worth of raw materials. The second product II requires $\frac{1}{4}$ hours of assembly, $\frac{1}{4}$ hours of testing, and 8 RMB worth of raw materials. The product of type III requires $\frac{1}{2}$ hours for assembly, $\frac{1}{4}$ hours of testing, and 30 RMB worth of raw materials. Given the current personnel of the company, there can be at most 150 hours of assembly labor and 80 hours of testing each day. Products of the first and second type have a market value of 70 RMB and 62 RMB, respectively. Furthermore, products of type III have a market value of 110 RMB.

a) Formulate a linear optimization problem that maximizes the daily profit of the company.

b) Consider the following modification to the original problem: Suppose that up to 60 hours of overtime assembly labor can be scheduled, at a cost of 50 RMB per hour. Can it be easily incorporated into the linear optimization formulation and how?

c) Solve the original LP in part a) using `MATLAB` or `Python` (e.g., via `CVX`).

Problem 3 (Scaffolding): (approx. *20* pts)
A construction company has to move scaffolding from three building sites (A, B, C) to two new building sites (1, 2). The scaffolds consist of iron rods. The table below shows the current number of iron rods at each building site and the number needed for the planned constructions:

Rods	A	B	C	1	2
Present	7000	6000	4000	0	500
Needed	0	2000	500	8000	6000

The following table provides the cost (in 0.1 RMB) of moving one iron rod between each pair of constructions sites:

Costs	A	B	C	1	2
A	0	5	3	9	6
B	5	0	9	7	4
C	3	9	0	4	6
1	9	7	4	0	8
2	6	4	6	8	0

Write down a linear optimization problem to compute the least costly way to move the scaffolding and rods such that the needs are met.

Solve the problem using software (e.g., via `CVX` and `MATLAB` or `Python`).

Problem 4 (Inheritance): (approx. *20* pts)

Unfortunately, Lady Dimitrescu has passed away. She has left the following legacy:

Item Description	No of Items	Value (per Item, in 10000$)	Comments
Sketch by Mondrian	1×	8	
Bust of Alexander the Great	1×	0.5	
Yuan Dynasty Chinese Vase	1×	3.5	
911 Porsche	1×	6	
Diamonds	3×	1.2	
Louis XV Sofa	1×	0.3	
Jack Russell Race Dogs	2×	0.3	The will mentions that the dogs can not be separated.
Ancient Sculpture	1×	1	
Sailing Boat	1×	2	
Harley Davidson Motorbike	1×	1	

Her legacy needs to be shared among her three daughters Bela, Daniela, and Cassandra.

The will mentioned that all items should be distributed equally among all daughters according to their (total) values. However, as the first-born daughter, Cassandra can receive a higher total share than Bela or Daniela. As the last-born daughter, Bela will receive the lowest share if an equal distribution of the values is not achievable.

a) Formulate an (integer) optimization problem to find a suitable partition of the items that allows to minimize the differences between the total values of Bela's and Cassandra's parts. Provide detailed explanations of your model!

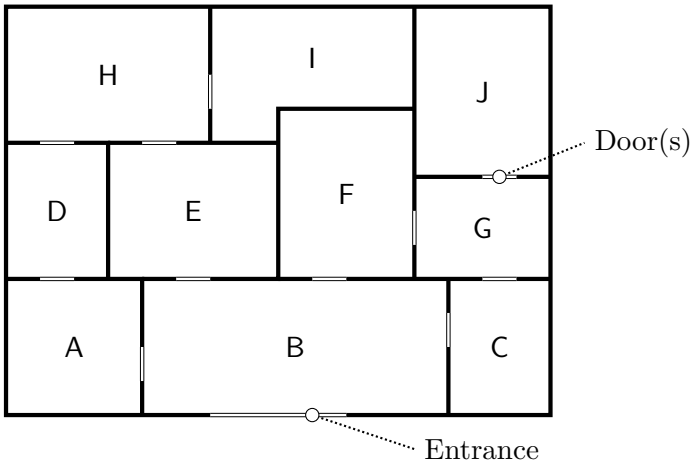
b) Is it possible to equally distribute the legacy among Bela, Daniela, and Cassandra?

Find an optimal solution to this inheritance problem (based on your formulation in part a)) and explain or discuss your choice in detail!

Problem 5 (Museum Guards): (approx. *20* pts)

A museum director must decide how many guards should be employed to control a newly opened wing of a museum. Budget cuts have forced the director to station guards at the doors between rooms — guarding two rooms at once.

A map of the wing and different rooms (A—J) is shown below:



Formulate an optimization problem to minimize the total number of required guards (while each of the rooms is controlled by at least one guard).

Describe and explain your optimization model briefly.

Information: `MATLAB` / `Python` Code

- For those questions that ask you to write `MATLAB` or `Python` code to solve the problem, please attach the code to the homework. Please also clearly state the optimal solution and the optimal value that you have obtained in your solution sheet. You do not need to attach the outputs in the command windows of `MATLAB` or `Python`.

Sheet 1 is due on **Feb, 14th**. Submit your solutions before **Feb, 14th**, **11:00 pm**.